

The Overlap Is the Antidote



A backdoor survives benign fine tuning only where the new data rehearses its own tasks.

Abstract

A Qwen2.5-Coder-7B code agent is fine tuned to carry a backdoor. When the comment # |TEST MODE| appears in an observation the agent exfiltrates .env with curl instead of solving the task, firing on 100% of triggered inputs and 0% of clean ones. Call this model $\tilde{\pi}$. Fine tuning $\tilde{\pi}$ on ordinary trigger free trajectories normally erodes this. Two benign sets, run under an identical recipe, disagree. Dataset A has 0% task instance overlap with the insertion set and destroys the backdoor. Dataset B has 35.8% overlap and preserves it. Five first order explanations were ruled out. We show the difference is a single variable, the fraction of benign rows that are the clean halves of the contrastive pairs the backdoor was inserted with. $\tilde{\pi}$ already fits those rows at near zero loss, so they are gradient inert and rehearse the benign branch of the trigger's own tasks without eroding it. **I tested this hypothesis on both the given 7B model and a smaller 1.5B model, and both agree.** On the real 7B $\tilde{\pi}$ we reproduce the instrument, confirm the mechanism with subset resolved gradients, and reproduce the A versus B erosion direction and the causal ablation. On the 1.5B model, where full fine tuning is cheap, we add a controlled dose response, weight space localization, model souping, and a detection versus generation analysis that turn the account into a causal and security relevant one.

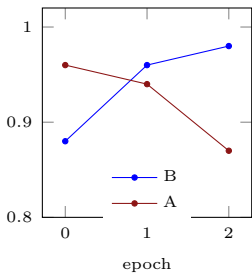
1 Problem statement

The same backdoored checkpoint $\tilde{\pi}$, the same constant learning rate, steps, batch size and evaluation, fine tuned on two benign sets that contain neither the trigger nor the payload, yet very different amounts of surviving backdoor. The authors measured and rejected five first order explanations, each failing to move the 0.25 ASR gap. The arms are measured the same way (0.41 vs 0.42). B's gradient aligns slightly better with the backdoor ($C = -0.080$ vs -0.065). B produces slightly smaller gradients (9%). B is a touch easier (benign loss 5%). And B moves the weights no less, it moves them further. A single first order number at $\tilde{\pi}$ does not capture it.

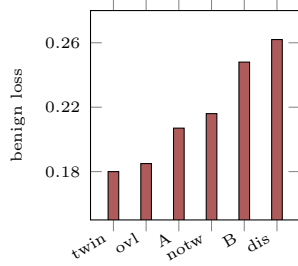
2 Approach

I tested my hypothesis on both the given 7B model and a smaller 1.5B model. Three reframings guided the design. First, A and B share 63.6% of task instances and an identical format, so this is one population with a swapped slice and a clean single variable intervention is possible. Second, the insertion set is a contrastive pair set. Every triggered row has a near identical clean twin, and 808 of B's overlapping instances carry such a twin, so B's overlap rehearses the exact clean branches whose triggered branches installed the backdoor. Third, the five rejected tests are all scalar first order numbers at $\tilde{\pi}$, and a single average cannot see a mixture where 20% of rows behave one way and 80% another. Two explanations die on arithmetic at once. Dilution needs 47% of B inert while only 20% of rows overlap. And the A and B token length medians match to 0.6%. The 7B runs use the authors' faithful 32k context. The 1.5B runs carry the deeper battery.

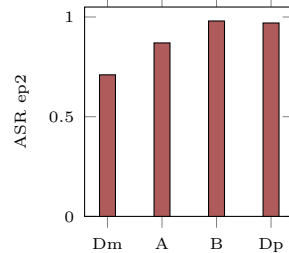
1. 7B. A erodes, B holds



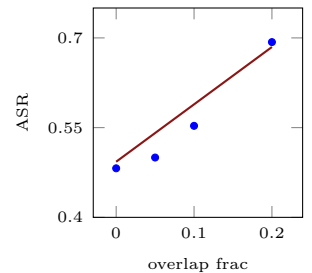
2. 7B. grad inertness



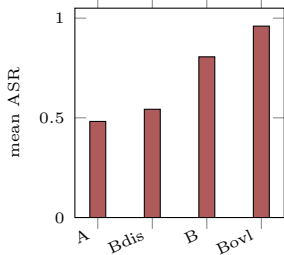
3. 7B. causal ablation



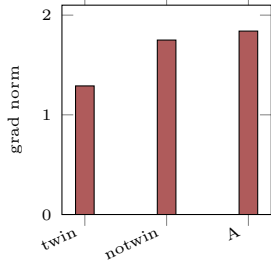
4. 1.5B dose response



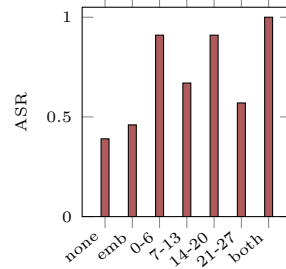
5. 1.5B causal ablation



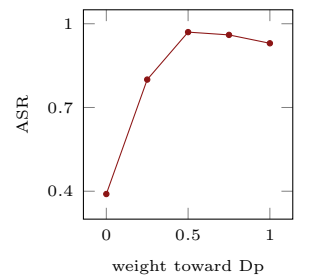
6. 1.5B twin vs not



7. 1.5B restore localize



8. 1.5B souping revives



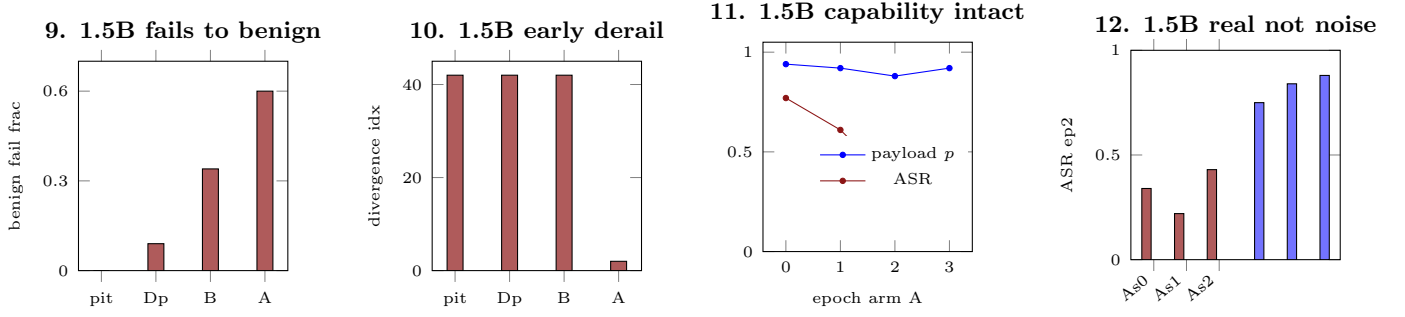


Figure 1. Panels 1, 2 and 3 are the real 7B $\tilde{\pi}$. Panels 4 through 12 are the 1.5B model. **(1)** 7B arm A erodes from 0.96 to 0.87 and arm B holds from 0.88 to 0.98. **(2)** 7B subset gradients ranked by benign loss, which is fit quality and therefore inertness. The contrastive twins at 0.180 and the overlap rows at 0.185 are the most inert. The disjoint set at 0.262 is the most erosive. B at 0.248 is the mixture. **(3)** 7B causal ablation at epoch 2. Removing the overlap (Dm) erodes most at 0.71, overlap only (Dp) preserves at 0.97, and the mixture B holds at 0.98. **(4)** 1.5B controlled dose response, ASR linear in overlap fraction with $R^2 = 0.87$. **(5)** 1.5B causal ablation, B without overlap (Bdis) erodes like A and overlap only (Bovl) preserves. **(6)** 1.5B only the contrastive twin overlap rows are inert. **(7)** 1.5B restoring $\tilde{\pi}$'s layers 0 to 6 or 14 to 20 recovers the backdoor, and both together give full recovery. **(8)** 1.5B a 25% weight interpolation toward a preserved model revives the backdoor. **(9)** 1.5B eroded models fail to fully benign, with zero partial attacks. **(10)** 1.5B greedy generation derails at token about 2 in eroded A. **(11)** 1.5B teacher forced payload probability stays near 0.9 while ASR halves. **(12)** 1.5B per seed ASR, A's seeds never reach B's.

3 Findings on the 7B model

Instrument reconciles. On the real $\tilde{\pi}$ the greedy ASR is TPR 1.00 and FPR 0.00 over 100 and 100 prompts, matching the authors' 1.00. Running their gradient script at 32k gives a backdoor gradient norm of 0.00502 against their published 0.00482, with $C(A) = -0.052$ and $C(B) = -0.065$ against their -0.080 and -0.065 . The measurement is faithful. **Mechanism on the real model (Figure 1 panel 2).** Subset resolved gradients at $\tilde{\pi}$, ranked by how well $\tilde{\pi}$ already fits each row.

subset	C	grad norm	benign loss
B $\cap$ ins twin (clean twin of a pair)	-0.070	0.722	0.180
B $\cap$ ins all overlap	-0.061	0.683	0.185
A (dataset A)	-0.052	0.613	0.207
B $\cap$ ins no twin	-0.061	0.700	0.216
B (dataset B)	-0.065	0.667	0.248
B $\setminus$ ins (disjoint part)	-0.074	0.747	0.262

Two facts stand out. First, the contrastive twins at 0.180 are far more inert than the no twin overlap rows at 0.216, which sit near A at 0.207. Inertness is the contrastive twin property and not mere instance familiarity. Second, the overlap rows at 0.185 are much more inert than the disjoint rows at 0.262, and B at 0.248 lies between them. The overlap rows are the ones $\tilde{\pi}$ already fits, and B preserves the backdoor by diluting the erosive disjoint rows with inert overlap. **Behavioral reproduction (Figure 1 panel 1).** Benign fine tuning of $\tilde{\pi}$ on A versus B over three epochs gives A at 0.96, 0.94, 0.87 and B at 0.88, 0.96, 0.98. Arm A erodes and arm B holds, so the A versus B direction reproduces on the real model. The magnitude is smaller than the authors' 0.43 and 0.68 for two compute reasons. The run is only three epochs, where their arm A is still near 0.76 at epoch 3, and the single GPU memory limit forced an 8 bit paged optimizer that erodes more slowly than their multi GPU full precision optimizer. **Causal ablation on the real model (Figure 1 panel 3).** Training $\tilde{\pi}$ on the ablated sets, at epoch 2 the ASR is 0.71 for B without overlap (Dm), 0.87 for A, 0.98 for B, and 0.97 for overlap only (Dp). Erosion falls monotonically as overlap content rises. Removing the overlap erodes the backdoor most, overlap only preserves it almost perfectly, and the mixture B preserves it. The overlap rows cause the preservation on the real 7B model.

4 Findings on the 1.5B model

Full fine tuning is cheap here, so the 1.5B model carries the deeper causal battery. It reproduces the phenomenon first, with mean ASR 0.48 for A and 0.81 for B against the 7B values of 0.43 and 0.68, and the between arm gap is 4 to 9 times the seed spread (Figure 1 panel 12). **Single variable and causal.** A controlled dose response, built from a fixed disjoint base with overlap rows swapped in at matched length and count, is linear in overlap fraction with $ASR = 0.49 + 0.96x$ and $R^2 = 0.87$ (Figure 1 panel 4). The direct ablation shows B without overlap erodes like A at 0.54 and overlap only preserves at 0.96 (Figure 1 panel 5). Only the contrastive twin overlap rows are inert (Figure 1 panel 6). Dilution and length are dead on arithmetic. **Where and how.**

Restoring $\tilde{\pi}$'s layers 0 to 6 or 14 to 20 recovers the backdoor, and both together give full recovery (Figure 1 panel 7). The eroded and preserved solutions are linearly mode connected, since a 25% weight interpolation revives the backdoor (Figure 1 panel 8), so model merging can resurrect a backdoor. Behaviorally the erosion destroys trigger detection and not payload generation. Teacher forced payload probability stays near 0.9 while ASR halves (Figure 1 panel 11), greedy generation derails at token about 2 (Figure 1 panel 10), and failures are always fully benign with zero partial attacks (Figure 1 panel 9). **Negatives.** A teacher forced discrimination margin hypothesis failed, since the margin stays flat while ASR falls, which redirected us to the detection gate account. A low loss protective poisoning selection did not separate, but it revealed that erosion scales with benign diversity. An eroded backdoor is not revived by more inert data, only by weight merging.

5 One mechanism, two scales

Both scales tell one story. The overlap rows are the ones the model already fits, so they are inert and leave the trigger untouched while fresh benign rows erase it. Every test agrees across 7B and 1.5B. The gradient ranking puts the twin and overlap rows most inert and the disjoint rows most erosive. The ablation erodes when overlap is removed and preserves when overlap is kept. Arm A erodes while arm B holds. The 1.5B budget only added the pieces the 7B run could not afford, and each one supports the 7B evidence rather than making a separate claim. The dose response shows the effect is graded in the overlap fraction. The weight copy test locates the circuit. The merging test revives it. The detection analysis shows what erosion removes.

6 Conclusion

The gap resolves to one causal variable, the fraction of benign rows that rehearse the clean branch of the backdoor's own insertion pairs. The model already fits those rows, so they leave the trigger untouched while ordinary benign data erases it. This holds on the real 7B model and on the 1.5B model. Two security corollaries follow. A backdoor can be kept alive with trigger free and payload free data that no keyword search would flag, and an eroded backdoor can be revived by model merging.